Esteban Nicolás Larena

Senior Full Stack Engineer | Java · Spring Boot · .NET · Angular

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SUMMARY

Senior Software Engineer with 5+ years building backend systems and end-to-end products across banking, insurance, energy, and retail. Strong background in Java/Spring Boot and .NET, with extensive full-stack experience in Angular/TypeScript (5+ years) and native Android in Kotlin. Work guided by SOLID principles, Clean Code, and Clean Architecture; comfortable with hexagonal architectures (Ports & Adapters), microservices, and monoliths as the context demands. Designed and built from scratch a .NET microservices system for YPF, shipping 10–15 features per sprint at 80% unit test coverage with SonarQube quality gates. Founder of FushiLin, a SaaS language-learning platform in production with dual-payment integration (Mercado Pago + Paddle) for local and international revenue. Experienced with legacy modernization (IBM i, Lansa, RPG, ASP Classic), BPM platforms (Engage at BBVA), and ETL processes (SSIS). Currently completing a CS degree with a thesis on LLM-based code generation. Bias for shipping, clean architecture, and direct communication with technical and business stakeholders.

EXPERIENCE

Full Stack Software Engineer

FlexTech (client: Banco Mariva) · Sep 2026 – Present · Argentina

- Development on banking systems (DB2, Java/Spring Boot, WSO2, Docker, Jenkins, Linux) for Banco Mariva.
- **Designed and built a RAG-based documentation assistant** over internal technical documentation (Spring Boot 3.4 / Java 21 + React + PostgreSQL 17 / pgvector, Python loader): hybrid search (pgvector + BM25/Lucene with RRF), cross-encoder reranking and HyDE, with embedding and reranking models run locally on ONNX. **Measured result: –92% tokens per query** with equal or better quality (+~22% on correctness, blind LLM judge), verifiable citations and near-zero hallucination; validated with a custom evaluation set (recall@k, MRR).

Tech: Java, Spring Boot, Python, React, TypeScript, PostgreSQL, pgvector, BM25/Lucene, DB2, ONNX, RAG, Claude, Claude Code, Claude Code Console, WSO2, Tomcat, Swagger, Docker, Jenkins, Linux, Git, GitLab.

Full Stack Software Engineer

IAM (Instituto Asegurador Mercantil) · Jun 2025 – Sep 2026 · Argentina

- Modernizing core insurance systems on IBM i (AS/400): analyzing and migrating RPG code to RDMLX, maintaining Lansa applications (VLF, RAMP), and building new services in Java/Spring Boot and .NET integrated with the legacy stack.
- Integrations with Argentine tax authorities (AFIP/ARCA) for regulatory compliance, including translating technical documentation for the accounting team to configure the company's fiscal user.
- Custom ETL processes for data integration, implemented with Lansa, Java, and JavaScript.
- Database integration across DB2 for i, SQL Server, MySQL, PostgreSQL, and Redis.
- Cross-platform server administration on Windows, Linux, and IBM i: service management (services.msc/sc, systemctl), IIS, SSH, RDP, FTP, scripting in PowerShell, Bash, and CL; QSH shell invocation from Lansa via system call.
- Maintaining and developing WordPress applications.
- Self-taught Lansa to a productive level in under 2 months despite scarce public documentation.
- Continuous technical mentoring of a developer on the team, focused on Lansa and Java.
- Daily integration of GenAI tools (Claude, Gemini, ChatGPT) into the development workflow.

Tech: Java, Spring Boot, .NET, JavaScript, HTML/CSS, Lansa, VLF, RAMP, RDMLX, DB2 for i, SQL Server, MySQL, PostgreSQL, Redis, Windows Server, IIS, Linux, IBM i, Mocha Software (TN5250 client), PowerShell, Bash, CL, QSH, WordPress, Maven, Gradle, NuGet, Git.

Founder & End-to-End Software Engineer

FushiLin · 2022 – Present · fushitelingua.com

- My own Japanese language-learning SaaS based on structured mnemonics, live in production. End-to-end development alongside my formal roles.
- **Backend:** Spring Boot + JPA + MariaDB / MySQL (Cloud SQL), automated migrations.
- **Frontend:** Angular + TypeScript. **Mobile:** Native Android in Kotlin (Google Play Billing); native iOS in Swift / SwiftUI (in development).
- **Infrastructure:** GCP Compute Engine, Nginx as multi-microservice reverse proxy, SSL/DNS management. **Authentication:** OAuth2 / JWT. **Build & CI/CD:** Maven, Gradle, GitLab Pipelines.
- **Payments (in production):** dual Mercado Pago (Argentina) + Paddle (international) integration, solving Argentine regulatory restrictions on cross-border revenue.
- **Game version (in development):** educational game in Unreal Engine (C++ and Blueprints), with a distributed architecture of Spring Boot and .NET microservices over SQL Server, integrated with FushiLin.

Tech: Java, Spring Boot, Angular, TypeScript, Kotlin, Swift/SwiftUI, C++ (Unreal Engine), .NET, MariaDB, MySQL, GCP, Nginx, Mercado Pago, Paddle, OAuth2, JWT, Maven, Gradle, GitLab Pipelines.

Freelance Software Engineer
Self-employed · 2022 – 2025 · Argentina · Remote

- JWT authentication service built in .NET + SQL Server for an independent client.
- WordPress sites for various small clients.

Backend Software Engineer
Tsoft S.A. · Jan 2022 – Jan 2024 · Argentina

- Built and maintained .NET (C#) and Java/Spring Boot applications for top Argentine enterprise clients: Banco Hipotecario (Spring Boot), YPF (.NET), and Sancor Seguros (.NET).
- **Designed and built from scratch a .NET microservices system for YPF**, which handled approval, decommissioning, and equipment-change requests for their service stations: sustained a delivery pace of 10–15 features per sprint with 80% unit test coverage and SonarQube quality gates. The architecture allowed shipping changes within the same sprint, leaving room for business-model analysis and cross-team support.
- Built complex Angular/TypeScript frontend features for YPF, unblocking the frontend team on advanced TypeScript, state management, and backend integration.
- Enterprise authentication with Azure AD/OpenID (OAuth2) and integrations with Microsoft Graph, Google, and Azure APIs.
- Integrations with SQL Server, MariaDB, Teradata, and Azure Blob Storage.
- Unit testing with JUnit + Mockito (Java) and NUnit (.NET), enforced through SonarQube quality gates.
- Monitoring nightly batch processes with Grafana, identifying and diagnosing occasional production errors.
- Resolved production incidents in Sancor Seguros' legacy .NET MVC system, with direct user communication. Performance improvements on heavy processes.
- Performed functional analysis across all assigned projects.

Tech: Java, Spring Boot, .NET, .NET MVC, Angular, TypeScript, Entity Framework, FluentValidation, LINQ, Swagger, SQL Server, MariaDB, Teradata, Azure (AD, Blob Storage, DevOps, Pipelines), OAuth2, JUnit, Mockito, NUnit, Maven, Gradle, NuGet, SonarLint, SonarQube, Postman, Jira, Grafana, Git. Gitflow, Scrum, Kanban, CI/CD.

Full Stack Software Engineer
Squad · Jun 2020 – Dec 2021 · Argentina

- Manpower model with parallel assignments to multiple projects.
- **Development:** Spring Boot microservices for Pay per TIC, .NET / ASP Classic web applications, and native Android apps in Java.
- **BPMS and ETL implementation at BBVA (6 months):** loan delinquency workflows on Engage (banking BPM platform), including configuration, bank-system integrations, and custom code development. ETL processes built with SSIS (legacy in Visual Studio 2008).
- Supported 2 trainees with algorithmic problem-solving for ~2 months, complementing the team's main mentor.
- One-on-one mentoring of a trainee for ~5 months, jointly building a native Android application (Android Studio, Java).
- 100% unit and integration test coverage using JUnit + Mockito (Java) and NUnit (.NET).
- Authentication based on Keycloak (OAuth2 / OpenID Connect).
- Requirements gathering and cost estimation.

Tech: Java, Spring Boot, .NET, VB.NET, Xamarin with C#, ASP.NET Framework, ASP Classic, native Android (Java), Engage (BPM), SSIS / Visual Studio 2008, SQL Server, MariaDB, Redis, MongoDB, Kafka, Keycloak, OAuth2, JUnit, Mockito, NUnit, Maven, Gradle, NuGet, Docker, Kubernetes, Jenkins, Postman, Newman, Bitbucket, GitLab, SonarLint, SonarQube. Gitflow, Scrum, CI/CD.

Teaching Assistant — Programming I & Introduction to Programming (Python & C)
National University of Luján · Mar 2022 – Mar 2023

- Assisted in teaching programming logic and problem-solving. The course covered Python at the start and transitioned to C toward the end, with Git used throughout.

Teaching Assistant — Programming I (Pascal)
National University of Luján · Aug 2019 – Jul 2021

- Assisted in teaching programming logic and problem-solving in Pascal.

FEATURED PROJECTS

KajiJDK — JDK in Rust from scratch · 2026 – Present
From-scratch reimplementation of the Java platform core in Rust: a JVM that runs real bytecode, an own generational GC, a JVMS-strict bytecode verifier, and green threads with monitors. javap byte-identical vs OpenJDK 25. Own javac compiler and professional multithreading in development. Includes KajiLibrary, an own standard library covering 100% of the JDK 25 public API surface (4,153 classes and 48,893 members across 233 packages): built with an LLM-assisted pipeline and validated by differential testing against OpenJDK using my own tooling (javap, javac, jdb, jlink and a fuzzer).

BoxBreaker (BBK) — RPG compiler with a custom IR · 2026 – Present
RPG (IBM i / AS/400) compiler built on a custom intermediate language (BBK), aiming to make the RPG language portable beyond IBM hardware. IntelliJ plugin published on the JetBrains Marketplace.

Own CRM · 2025 – Present
Building a CRM system from scratch (in progress).

EDUCATION

B.Sc. in Information Systems — National University of Luján, Argentina

2017 – Present · all coursework completed. Thesis in progress on GenAI / LLM-based code generation for enterprise programming tasks, expected December 2026.

Associate Degree in Software Engineering (intermediate degree) — UNLu, 2017 – 2024.

Law (paused) — University of Buenos Aires · 2012 – 2016.

TECHNICAL SKILLS

Design principles & architecture: SOLID, Clean Code, Clean Architecture, Hexagonal Architecture (Ports & Adapters), microservices, monolithic, client/server.

Backend: Spring Boot, .NET, .NET MVC, Entity Framework, JPA/Hibernate, microservices, REST APIs.

Frontend: Angular + TypeScript (5+ years, productive and mentoring), HTML5, CSS, Bootstrap, JavaScript, jQuery. React: basic knowledge.

Mobile: Native Android (Kotlin on FushiLin, Java on Squad), Xamarin, native iOS (Swift / SwiftUI on FushiLin).

Languages: Java, Kotlin, Swift, C#, TypeScript, JavaScript, Python, PHP, C/C++, VBA, RDMLX, SQL, PowerShell, Bash, CL.

Databases: SQL Server, PostgreSQL, MySQL, MariaDB, DB2 for i, MongoDB, Redis, Teradata.

Cloud: GCP (Compute Engine, Cloud SQL), Azure (AD, Blob Storage, DevOps, Pipelines), Cloudflare (DNS, email routing, SSL/TLS), AWS (basic).

DevOps & Sysadmin: Docker, Kubernetes, Nginx, Linux, Windows Server, IIS, service management (systemctl, services.msc/sc), SSH, RDP, FTP, FortiClient VPN, Jenkins, GitHub Actions, GitLab Pipelines, Azure Pipelines, CI/CD.

Messaging & events: Kafka, RabbitMQ.

Security & identity: OAuth2, OpenID Connect, Azure AD, JWT, Keycloak, Microsoft Graph API.

BPM & ETL: Engage (loan delinquency workflows at BBVA), SSIS / Visual Studio 2008.

Payments: Mercado Pago, Paddle, Google Play Billing.

Government integrations: AFIP/ARCA (Argentine tax authorities).

GenAI in development workflow: Claude, Gemini, ChatGPT — daily productive use for code generation, assisted code review, debugging, and documentation. CS thesis on the topic.

Build tools: Maven, Gradle, NuGet.

Quality & testing: JUnit, Mockito, NUnit, SonarQube, SonarLint, TDD, unit and integration testing, Postman, Newman, Selenium, Swagger.

Legacy environments: IBM i (AS/400), Lansa, VLF, RAMP, RPG, ASP Classic, Mocha Software (TN5250 client).

Methodologies: Scrum, Kanban, Agile, Scrumban, Gitflow, CI/CD, continuous improvement.

LANGUAGES

Spanish: native.

English: advanced reading and writing, intermediate spoken.

Japanese: 6 months of cultural immersion living in Japan. JLPT N4 (Dec 2025), preparing for N3.

CONTINUING EDUCATION

- Continuous Development, Testing & Production Workflow (WFC) — Microways, 2021
- User-Centered Interactive Systems Design (UX) — Argentine Congress of Computer Science, 2020
- Professional Microservices Development — MUG, 2020
- TDD, Hibernate, and Spring — UNLu, 2019
- Training in basic cybersecurity, software development, and soft skills provided by client companies